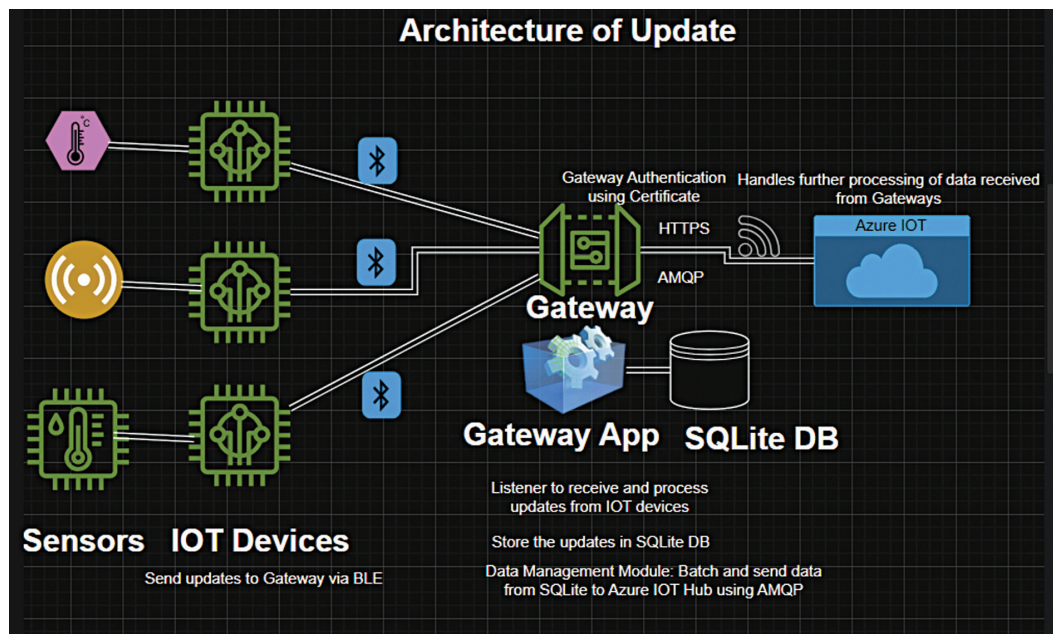


IoT GATEWAYS: A Layer Between IoT Devices And Cloud

Behind every smooth IoT experience, the gateway is doing the heavy lifting. It is not just plumbing for data; it can mean the difference between chaos and clarity in connected systems.



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In large-scale IoT setup scenarios, it is difficult to send data from IoT devices directly to the cloud as the network environment is not always connected to the internet, and protocols besides TCP/IP may govern IoT device operations. This is where the IoT gateway comes in, allowing data from multiple IoT devices or other embedded devices using different protocols to gather in one place before being sent to the cloud. IoT gateways are more than just a piece of network hardware; they serve as the control point that ensures data flows securely, efficiently, and in a business-ready format.

Importance of IoT gateway

In any IoT ecosystem, the gateway is as critical as the devices themselves. It acts as a central hub, managing communication between multiple devices, consolidating data, and ensuring secure transmission to the cloud.

Its core functions include connecting devices that use different communication protocols and translating them into formats the cloud can accept. For instance, a device operating on the Zigbee protocol cannot send data directly to the cloud; it must first connect to the gateway via Zigbee, which will then transfer the data to the cloud.

The same applies to Bluetooth low energy (BLE) devices. These can only operate within a Bluetooth network, so they connect to the gateway via BLE, which then transmits their data to the cloud.

Case study: Vibration monitoring in elevators located in airports

In many large facilities, security policies and environmental conditions dictate how connected devices can operate. For example, the use of Wi-Fi is restricted in airports.

In this case, the project involved